# Averages & Weighted Averages

Topic Test — Solutions & Answers

### Question 1

If a certain amount of money is divided equally among n persons, each one receives Rs 352. However, if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receive less than or equal to Rs 330. Then, the maximum possible value of n is

**[ Type in the Answer ] Correct Answer: 16**

**Solution:**

### Question 2

There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by 2x kg. If the weight of E is 12 kg more than that of D, then the value of x is

- A) 2
- B) 0.5
- **✓ C) 1 ← Correct Answer**
- D) 1.5

**Correct Answer: C**

**Solution:**

### Question 3

The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where $u=(x+y)/2$ and $v=(y+z)/2$. If $x \geq z$, then the minimum possible value of x is

**[ Type in the Answer ] Correct Answer: 105**

**Solution:**

**Step 1: Formulate equations from the given means.**

Mean of x, y, z is 80: $(x + y + z) / 3 = 80 \rightarrow x + y + z = 240$ ...(1)

Mean of x, y, z, u, v is 75: $(x + y + z + u + v) / 5 = 75$

Substitute (1): $(240 + u + v) / 5 = 75 \rightarrow u + v = 375 - 240 = 135$ ...(2)

**Step 2: Use the definitions of u and v.**

$u = (x+y)/2$ and $v = (y+z)/2$. Substitute into equation (2):

$(x+y)/2 + (y+z)/2 = 135 \rightarrow x + y + y + z = 270 \rightarrow x + 2y + z = 270$ ...(3)

**Step 3: Solve the system of equations for x, y, z.**

Subtract (1) from (3): $(x + 2y + z) - (x + y + z) = 270 - 240 \rightarrow y = 30$.

Substitute $y = 30$ back into (1): $x + 30 + z = 240 \rightarrow x + z = 210$.

**Step 4: Find the minimum value of x.**

We are given the condition $x \geq z$. From $x + z = 210$, we have $z = 210 - x$.

The condition $x \geq z$ becomes $x \geq 210 - x \rightarrow 2x \geq 210 \rightarrow x \geq 105$.

The minimum possible value of x is 105.

### Question 4

The strength of a salt solution is p% if 100 ml of the solution contains p grams of salt. If three salt solutions A, B, C are mixed in the proportion 1:2:3, then the resulting solution has strength 20%. If instead the proportion is 3:2:1, then the resulting solution has strength 30%. A fourth solution, D, is produced by mixing B and C in the ratio 2:7. The ratio of the strength of D to that of A is

- A) 3:10
- **✓ B) 1:3 ← Correct Answer**
- C) 1:4
- D) 2:5

**Correct Answer: B**

**Solution:**

**Case 1: Mixed in proportion 1:2:3, result is 20%.**

$(1a + 2b + 3c) / (1 + 2 + 3) = 20 \rightarrow a + 2b + 3c = 120$ ...(1)

**Case 2: Mixed in proportion 3:2:1, result is 30%.**

$(3a + 2b + 1c) / (3 + 2 + 1) = 30 \rightarrow 3a + 2b + c = 180$ ...(2)

**Step 1: Solve the system of linear equations.**

Subtract equation (1) from (2): $(3a + 2b + c) - (a + 2b + 3c) = 180 - 120$

$2a - 2c = 60 \rightarrow a - c = 30 \rightarrow a = c + 30$ ...(3)

**Step 2: Find the strength of solution D.**

Solution D is a mixture of B and C in the ratio 2:7.

Strength of D = $d = (2b + 7c) / (2 + 7) = (2b + 7c) / 9$.

Substitute $a = c + 30$ into equation (1): $(c+30) + 2b + 3c = 120$

$4c + 2b + 30 = 120 \rightarrow 2b = 90 - 4c \rightarrow b = 45 - 2c$.

Now substitute 'b' into the expression for d:

$d = (2(45 - 2c) + 7c) / 9 = (90 - 4c + 7c) / 9 = (90 + 3c) / 9 = (30 + c) / 3$.

**Step 3: Find the ratio d / a.**

$d = (30 + c) / 3$ and $a = c + 30$.

Ratio = $d / a = [(30 + c) / 3] / (c + 30) = 1/3$. The ratio is 1:3.

## Question 5

Onion is sold for 5 consecutive months at the rate of Rs 10, 20, 25, 25, and 50 per kg, respectively. A family spends a fixed amount of money on onion for each of the first three months, and then spends half that amount on onion for each of the next two months. The average expense for onion, in rupees per kg, for the family over these 5 months is closest to

**[ Type in the Answer ] Correct Answer: 18**

**Solution:**

**1. Assume a convenient value for the money spent.**

Let the fixed amount spent in each of the first three months be X. The amount spent in each of the next two months is X/2.

Choose X as the LCM of 10, 20, and 25: LCM(10, 20, 25) = 100. So X = Rs. 100, X/2 = Rs. 50.

**2. Calculate the quantity of onion purchased each month.**

Month 1: 100/10 = 10 kg. Month 2: 100/20 = 5 kg. Month 3: 100/25 = 4 kg.

Month 4: 50/25 = 2 kg. Month 5: 50/50 = 1 kg.

**3. Calculate the total quantity and total expense.**

Total quantity = 10 + 5 + 4 + 2 + 1 = 22 kg.

Total expense = 100 + 100 + 100 + 50 + 50 = Rs. 400.

**4. Calculate the average expense per kg.**

Average expense = Total Expense / Total Quantity = 400 / 22 = 200/11 ≈ 18.18.

The closest integer to 18.18 is 18.

## Question 6

Suppose hospital A admitted 21 less Covid infected patients than hospital B, and all eventually recovered. The sum of recovery days for patients in hospitals A and B were 200 and 152, respectively. If the average recovery days for patients admitted in hospital A was 3 more than the average in hospital B, then the number admitted in hospital A was

**[ Type in the Answer ] Correct Answer: 35**

**Solution:**

**1. Define variables.**

Let the number of patients in hospital B be x. Then the number of patients in hospital A is x - 21.

**2. Express average recovery days.**

Average days for A (Avg_A) = 200 / (x - 21). Average days for B (Avg_B) = 152 / x.

**3. Set up the equation based on the given condition.**

Avg_A = Avg_B + 3 → 200 / (x - 21) = (152 / x) + 3.

**4. Solve the equation for x.**

$200x = (x - 21)(152 + 3x) = 152x + 3x^2 - 3192 - 63x$

$200x = 152x + 3x^2 - 3192 - 63x \rightarrow 3x^2 - 111x - 3192 = 0$

Divide by 3: $x^2 - 37x - 1064 = 0$.

5. Factoring gives (x - 56)(x + 19) = 0. Since x cannot be negative, x = 56.

**6. Find the number of patients in hospital A.**

Number of patients in hospital A = x - 21 = 56 - 21 = 35.

### Question 7

The mean of all 4-digit even natural numbers of the form 'aabb', where a > 0, is:

- A) 4466
- B) 5050
- C) 4864
- **✓ D) 5544 ← Correct Answer**

**Correct Answer: D**

**Solution:**

**Step 1: Sum of '1100a' terms.**

For each of the 9 values of 'a', 'b' can be any of the 5 even digits. Each '1100a' term is counted 5 times.

Sum(a) = 5 × [1100(1) + 1100(2) + ... + 1100(9)] = 5 × 1100 × (1+2+...+9) = 5500 × 45 = 247,500.

**Step 2: Sum of '11b' terms.**

For each of the 5 values of 'b', 'a' can be any of the 9 digits from 1 to 9. Each '11b' term is counted 9 times.

Sum(b) = 9 × 11 × (0 + 2 + 4 + 6 + 8) = 99 × 20 = 1,980.

**Step 3: Total Sum and Mean.**

Total Sum = 247,500 + 1,980 = 249,480.

Mean = Total Sum / Number of terms = 249,480 / 45 = 5544.

### Question 8

The arithmetic mean of all the distinct numbers that can be obtained by rearranging the digits in 1421, including itself, is

- **✓ 2222) 2222 ← Correct Answer**
- 2442) 2442
- 2592) 2592
- 3333) 3333

**Correct Answer: 2222**

**Solution:**

### Question 9

A company has 40 employees. In 2022, the average bonus of the first 30 employees in the list was Rs. 40,000, the average bonus of the last 30 employees was Rs. 60,000, and the average bonus of the first 10 and last 10 employees together was Rs. 50,000. In 2023, the bonus of the first 10 employees increased by 100%, the bonus of the last 10 employees increased by 200%, and the bonus of the remaining 20 employees was unchanged. What was the average bonus, in rupees, of all 40 employees in 2023?

**[ Type in the Answer ] Correct Answer: 95000**

**Solution:**

1. **Define Groups:**

   Let A, B, C, D be the sum of bonuses for employees 1-10, 11-20, 21-30, and 31-40, respectively.

2. **Formulate Equations from 2022 Data:**

   First 30: (A+B+C)/30 = 40000 → A+B+C = 1,200,000.

   Last 30: (B+C+D)/30 = 60000 → B+C+D = 1,800,000.

   First 10 & Last 10: (A+D)/20 = 50000 → A+D = 1,000,000.

3. **Solve the System:**

   Subtracting the first two equations: (B+C+D) - (A+B+C) = 1.8M - 1.2M → D - A = 600,000.

   We have A+D = 1,000,000 and D-A = 600,000. Adding: 2D = 1,600,000 → D = 800,000.

   Then A = 1,000,000 - 800,000 = 200,000.

   The middle 20 employees' bonus sum is B+C = 1,200,000 - A = 1,000,000.

4. **Calculate 2023 Total Bonus:**

   New A (100% increase) = 200,000 × 2 = 400,000.

   New D (200% increase) = 800,000 × 3 = 2,400,000.

   B+C (unchanged) = 1,000,000.

   Total 2023 bonus = 400,000 + 1,000,000 + 2,400,000 = 3,800,000.

5. **Calculate 2023 Average Bonus:**

   Average = Total bonus / 40 = 3,800,000 / 40 = 95,000.

## Question 10

Consider six distinct natural numbers such that the average of the two smallest numbers is 14, and the average of the two largest numbers is 28. Then, the maximum possible value of the average of these six numbers is

- 23) 23
- 24) 24
- 23.5) 23.5
- **✓ 22.5) 22.5 ← Correct Answer**

**Correct Answer: 22.5**

**Solution:**

## Question 11

In an examination, the average marks of students in sections A and B are 32 and 60, respectively. The number of students in section A is 10 less than that in section B. If the average marks of all the students across both the sections combined is an integer, then the difference between the maximum and minimum possible number of students in section A is

**[ Type in the Answer ] Correct Answer: 63**

**Solution:**

**Possible values for 'a':**

$a+5=7 \rightarrow a=2$ (Minimum)

$a+5=10 \rightarrow a=5$

$a+5=14 \rightarrow a=9$

$a+5=35 \rightarrow a=30$

$a+5=70 \rightarrow a=65$ (Maximum)

## Question 12

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

**[ Type in the Answer ] Correct Answer: 70**

**Solution:**

**After the changes:**

The new numbers are $(x+7)$, $y$, and $(z-10)$. The new sum = $x+y+z - 3 = 84 - 3 = 81$.

The new mean is $81/3 = 27$.

The new mean (27) is 2 more than the middle number ($y$): $27 = y + 2 \rightarrow y = 25$.

**Use the difference between the new numbers:**

The difference between the new largest and smallest numbers is 64.

$(z - 10) - (x + 7) = 64 \rightarrow z - x - 17 = 64 \rightarrow z - x = 81$ ...(2)

**Solve for x and z:**

Substitute $y = 25$ into equation (1): $x + 25 + z = 84 \rightarrow x + z = 59$ ...(3)

From (2) and (3): $z - x = 81$ and $z + x = 59$.

Adding: $2z = 140 \rightarrow z = 70$. The largest number is 70.

## Question 13

Five students, including Amit, appear for an examination in which possible marks are integers between 0 and 50, both inclusive. The average marks for all the students is 38 and exactly three students got more than 32. If no two students got the same marks and Amit got the least marks among the five students, then the difference between the highest and lowest possible marks of Amit is

- 22) 22
- 21) 21
- 24) 24
- **✓ 20) 20 ← Correct Answer**

**Correct Answer: 20**

**Solution:**

**To find the lowest possible marks for Amit (M5_min):**

To minimize M5, we must maximize the scores of the other four students.

M1 = 50, M2 = 49, M3 = 48 (highest distinct scores ≤ 50 and > 32).

M4 = 32 (highest possible score ≤ 32). Sum of these four = 50+49+48+32 = 179.

M5_min = Total Marks - Sum = 190 - 179 = 11.

**To find the highest possible marks for Amit (M5_max):**

To maximize M5, we must minimize the scores of the other four students.

M1, M2, M3 must be the smallest distinct integers > 32: M3=33, M2=34, M1=35.

M4 must be > M5 and ≤ 32. Let M5 = x; M4 must be at most 31 (since M5 < M4 ≤ 32).

If M5 = 31, then M4 must be > 31 and ≤ 32, so M4 = 32.

Sum of other three: M1 + M2 + M3 = 190 - (32 + 31) - 63 = 190 - 63 = 127.

Smallest distinct integers M1 > M2 > M3 > 32 summing to 127: e.g. 33, 44, 50. Yes, this works.

So M5_max = 31. Difference = M5_max - M5_min = 31 - 11 = 20.

### Question 14

Manu earns ■4000 per month and wants to save an average of ■550 per month in a year. In the first nine months, his monthly expense was ■3500, and he foresees that, tenth month onward, his monthly expense will increase to ■3700. In order to meet his yearly savings target, his monthly earnings, in rupees, from the tenth month onward should be

- **✓ 4400) 4400 ← Correct Answer**
- 4200) 4200
- 4300) 4300
- 4350) 4350

**Correct Answer: 4400**

**Solution:**

**For the first 9 months:**

Monthly earnings = ■4000. Monthly expense = ■3500. Monthly savings = 4000 - 3500 = ■500.

Total savings in first 9 months = $9 \times 500$ = ■4500.

**For the last 3 months (10th, 11th, 12th):**

Remaining savings required = Total target - Savings made = $6600 - 4500$ = ■2100.

Required monthly saving for the last 3 months = $2100 / 3$ = ■700.

**From the 10th month onward:**

Monthly expense = ■3700. Let the new monthly earnings be $E$.

Monthly Savings = $E$ - Monthly Expense → $700 = E - 3700$ → $E$ = ■4400.

### Question 15

In a football tournament, a player has played a certain number of matches, and 10 more matches are to be played. If he scores a total of one goal over the next 10 matches, his overall average will be 0.15 goals per match. On the other hand, if he scores a total of two goals over the next 10 matches, his overall average will be 0.2 goals per match. The number of matches he has played so far is:

**[ Type in the Answer ] Correct Answer: 10**

**Solution:**

**Scenario 1: 1 goal in the next 10 matches gives an average of 0.15.**

Total goals = $g + 1$. Total matches = $n + 10$.

$(g + 1) / (n + 10) = 0.15 \rightarrow g + 1 = 0.15n + 1.5 \rightarrow g = 0.15n + 0.5$ ...(Eq. 1)

**Scenario 2: 2 goals in the next 10 matches gives an average of 0.2.**

Total goals = $g + 2$. Total matches = $n + 10$.

$(g + 2) / (n + 10) = 0.2 \rightarrow g + 2 = 0.2n + 2 \rightarrow g = 0.2n$ ...(Eq. 2)

$0.15n + 0.5 = 0.2n \rightarrow 0.5 = 0.05n \rightarrow n = 0.5 / 0.05 = 10$.

### Question 16

The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

**[ Type in the Answer ] Correct Answer: 92**

**Solution:**

### Question 17

One part of a hostel's monthly expenses is fixed, and the other part is proportional to the number of its boarders. The hostel collects ■1600 per month from each boarder. When the number of boarders is 50, the profit of the hostel is ■200 per boarder, and when the number of boarders is 75, the profit of the hostel is ■250 per boarder. When the number of boarders is 80, the total profit of the hostel, in INR, will be

20200) 20200

✓ **20500) 20500 ← Correct Answer**

20800) 20800

20000) 20000

**Correct Answer: 20500**

**Solution:**

**Case 1: N=50, Profit/boarder = 200**

200 = 1600 - k - F/50 → k + F/50 = 1400 ...(1)

**Case 2: N=75, Profit/boarder = 250**

250 = 1600 - k - F/75 → k + F/75 = 1350 ...(2)

**Subtract (2) from (1):**

(F/50) - (F/75) = 50 → (3F - 2F) / 150 = 50 → F = 7500.

Substitute F in (1): k + 7500/50 = 1400 → k + 150 = 1400 → k = 1250.

**For N=80:**

Profit = R - E = 1600N - (F + kN) = 1600(80) - (7500 + 1250×80)

Profit = 128000 - (7500 + 100000) = 128000 - 107500 = 20500.

## Question 18

The average of three integers is 13. When a natural number n is included, the average of these four integers is an odd integer. The minimum possible value of n is:

A) 3

B) 4

✓ **C) 5 ← Correct Answer**

D) 1

**Correct Answer: C**

**Solution:**

## Question 19

The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is:

A) 1:4

B) 1:2

✓ C) 4:1 ← Correct Answer

D) 3:1

**Correct Answer: C**

**Solution:**

**Using the weighted average formula:**

$w + 0.6 = (n \times w + m \times (w + 3)) / (n + m)$

$(w + 0.6)(n + m) = nw + m(w + 3)$

$wn + wm + 0.6n + 0.6m = nw + wm + 3m$

$0.6n + 0.6m = 3m \rightarrow 0.6n = 2.4m \rightarrow n/m = 2.4/0.6 = 4/1.$

## Question 20

The average marks of 4 girls and 6 boys is 24. Each of the girls has the same marks while each of the boys has the same marks. If the marks of any girl is at most double the marks of any boy, but not less than the marks of any boy, then the number of possible distinct integer values of the total marks of 2 girls and 6 boys is

✓ 21) 21 ← Correct Answer

20) 20

22) 22

19) 19

**Correct Answer: 21**

**Solution:**

**1. Formulate equations:**

Let a girl's marks be g and a boy's marks be b.

Average marks: $(4g + 6b) / 10 = 24 \rightarrow 4g + 6b = 240 \rightarrow 2g + 3b = 120.$

Condition on marks: $b \leq g \leq 2b.$

**2. Define the target variable T:**

We need the number of integer values for $T = 2g + 6b.$

From $2g + 3b = 120$, we can write $6b = 2(3b) = 2(120 - 2g) = 240 - 4g.$

Substitute into T: $T = 2g + (240 - 4g) = 240 - 2g.$

**3. Find the range of g:**

From $2g + 3b = 120$: $b = (120 - 2g)/3$. Substitute into the inequality:

a) $b \leq g \rightarrow (120\text{-}2g)/3 \leq g \rightarrow 120\text{-}2g \leq 3g \rightarrow 120 \leq 5g \rightarrow g \geq 24.$

b) $g \leq 2b \rightarrow g \leq 2(120\text{-}2g)/3 \rightarrow 3g \leq 240 - 4g \rightarrow 7g \leq 240 \rightarrow g \leq 240/7 \approx 34.28.$

So, the range for g is $24 \leq g \leq 240/7.$

**4. Find the range of T:**

$T = 240 - 2g$ is a decreasing function of g.

Max T (at min g=24): $T = 240 - 2(24) = 192.$

Min T (at max g=240/7): T = 240 - 2(240/7) = 240(1 - 2/7) = 1200/7 ≈ 171.43.

The range of T is [171.43, 192], so integer values are 172, 173, ..., 192.

**5. Count the integer values:**

Number of values = 192 - 172 + 1 = 21.